

FOURTH ITERATION

$$m = 1.34$$

$$b = 1.916$$

$$\text{Points } (1, 3) \quad (3, 6)$$

$$\alpha = 0.1$$

$$\text{for } (1, 3) \Rightarrow \hat{y}_1 = 1.34(1) + 1.916 = 3.256$$

$$\text{for } (3, 6) \Rightarrow \hat{y}_2 = 1.34(3) + 1.916 = 5.936$$

Predicted values (3.256, 5.936)

$$\text{Error} \Rightarrow \begin{cases} 3 - 3.256 = -0.256 \\ 6 - 5.936 = 0.064 \end{cases}$$

$$[-0.256, 0.064]$$

$$\frac{\Delta E}{\Delta m} = -\frac{2}{n} \sum (y_i - \hat{y}_i) x_i \Rightarrow \frac{-2}{2} (1(-0.256) + 3(0.064))$$

$$= -1(-0.064) = 0.064$$

$$\frac{\Delta E}{\Delta b} = \frac{-2}{2} (-0.256 + 0.064)$$

$$= -1(-0.192)$$

$$= 0.192$$

Updated m and b

$$m_{\text{new}} = 1.34 - (0.1(0.064)) = \underline{\underline{1.3336}}$$

$$b_{\text{new}} = 1.916 - (0.1(0.192)) = \underline{\underline{1.8968}}$$